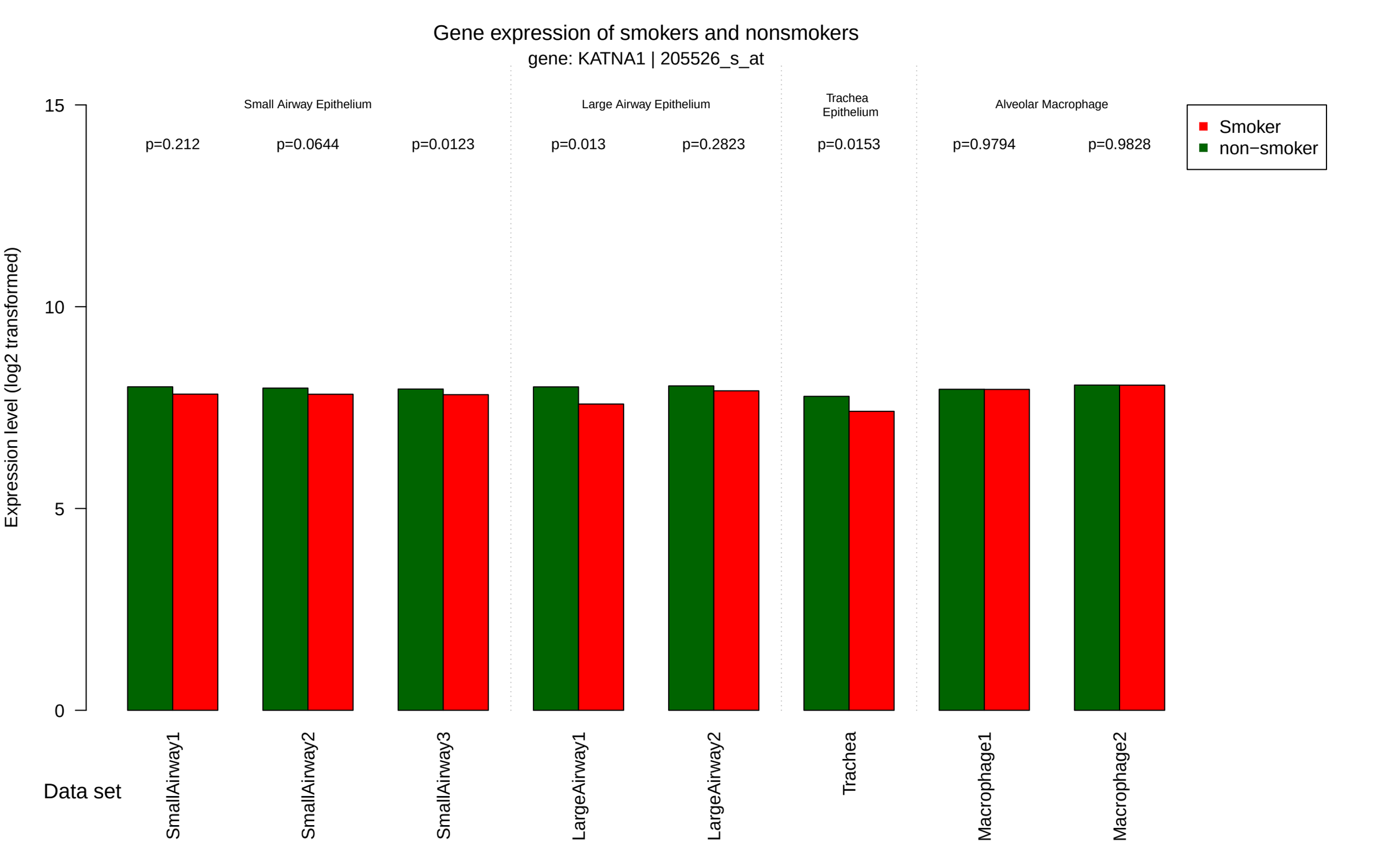
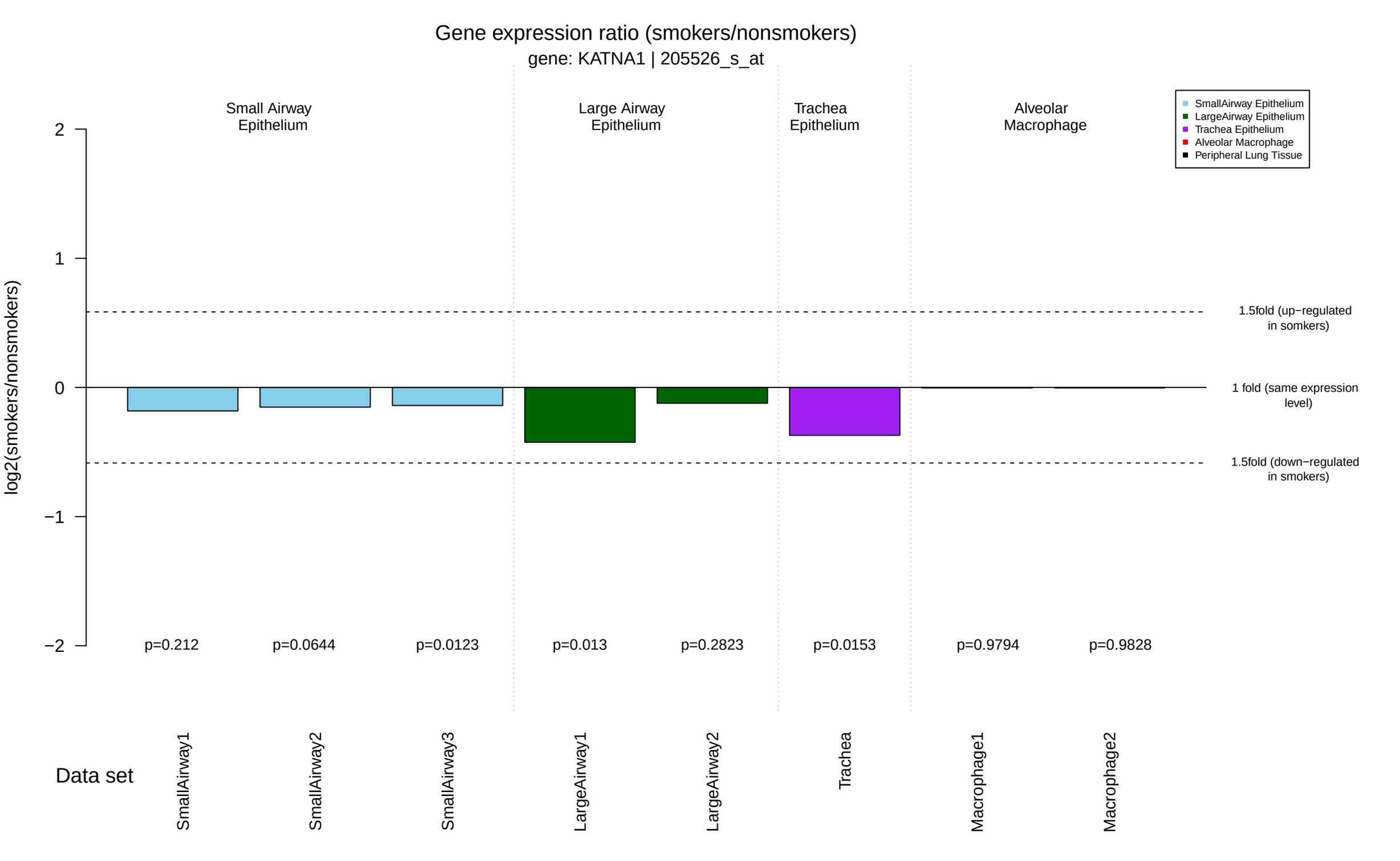
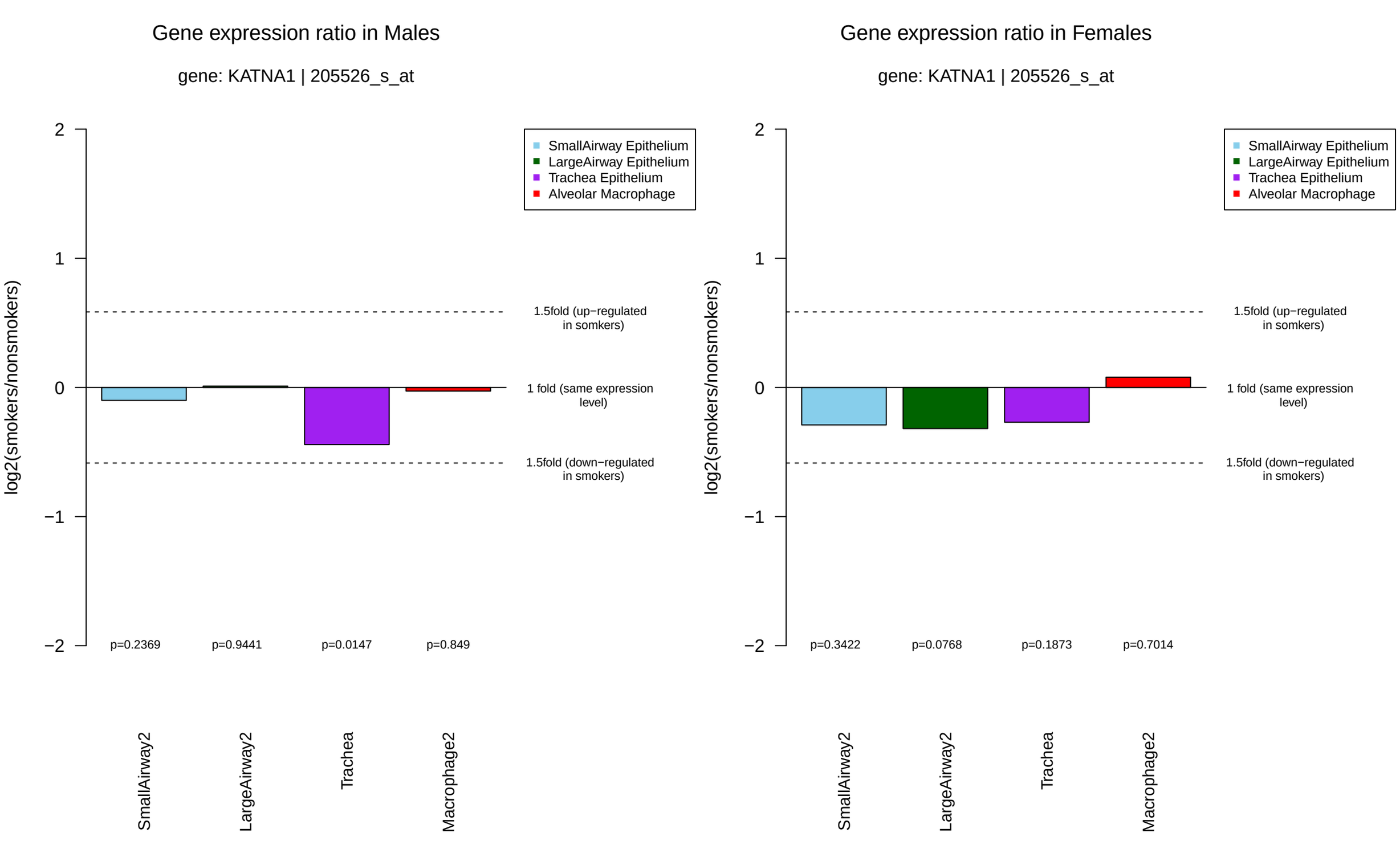
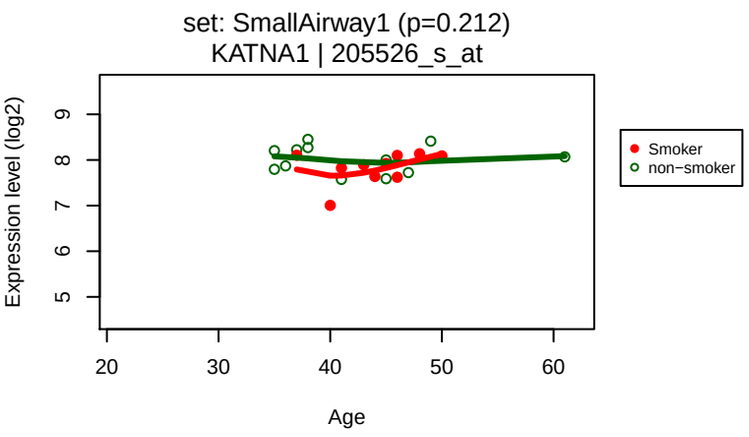
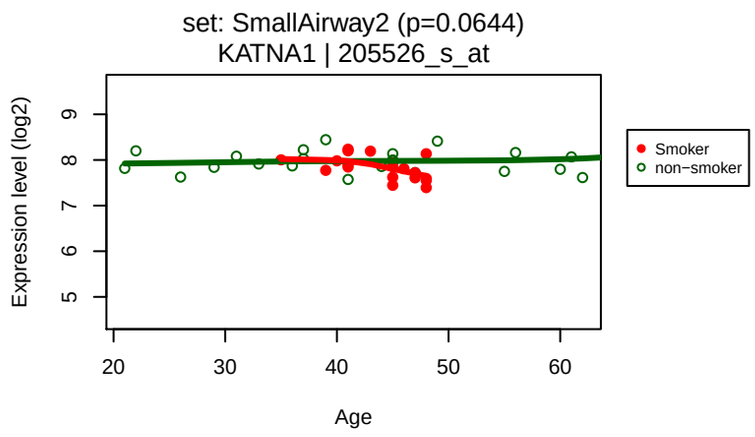
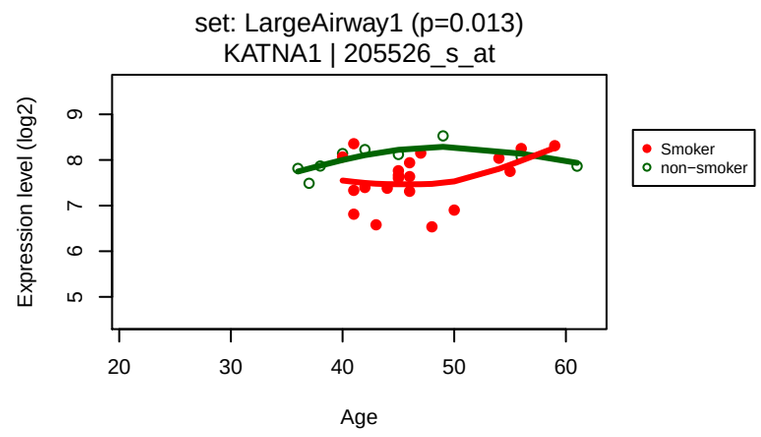

Gene expression of smokers and nonsmokers
gene: KATNA1 | 205526_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.212
p=0.0644
p=0.0123
p=0.013
p=0.2823
p=0.0153
p=0.9794
p=0.9828
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

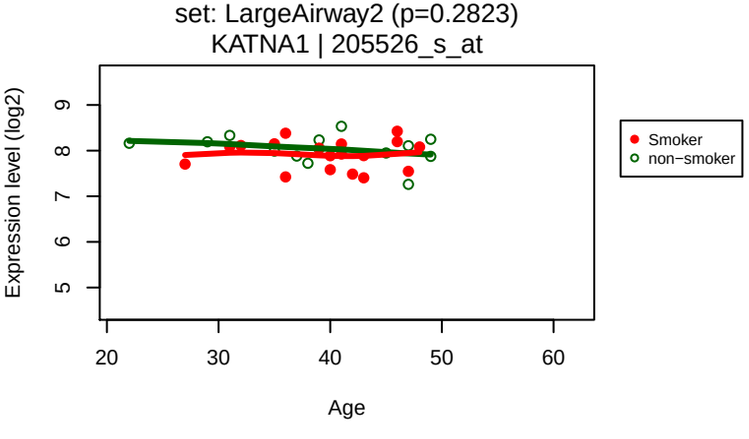

Gene expression ratio (smokers/nonsmokers)
gene: KATNA1 | 205526_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.212
p=0.0644
p=0.0123
p=0.013
p=0.2823
p=0.0153
p=0.9794
p=0.9828
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

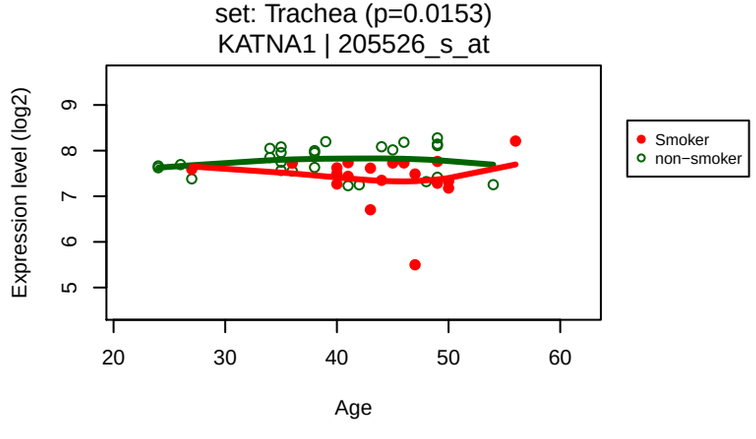

Gene expression ratio in Males
gene: KATNA1 | 205526_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2369
p=0.9441
p=0.0147
p=0.849
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KATNA1 | 205526_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3422
p=0.0768
p=0.1873
p=0.7014
SmallAirway2
LargeAirway2
Trachea
Macrophage2

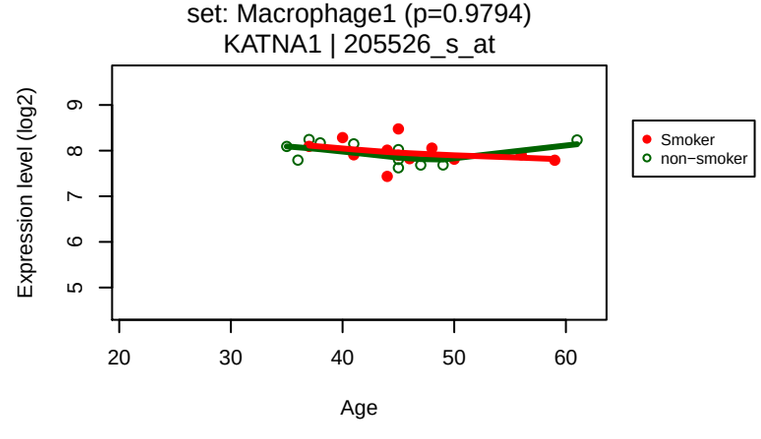

set: SmallAirway1 (p=0.212)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker

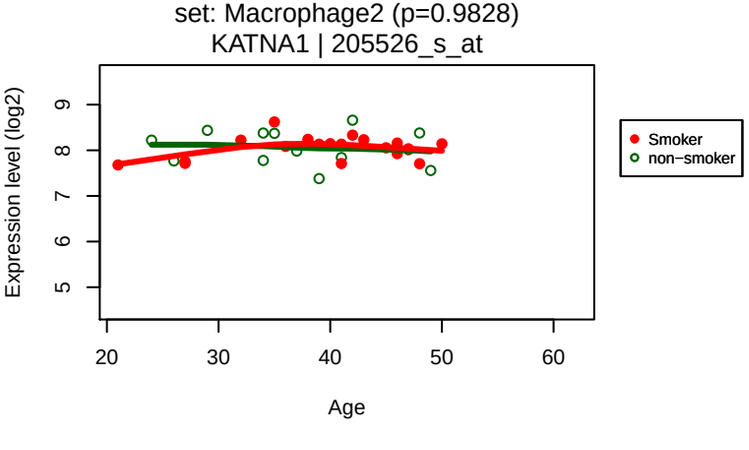

set: SmallAirway2 (p=0.0644)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker

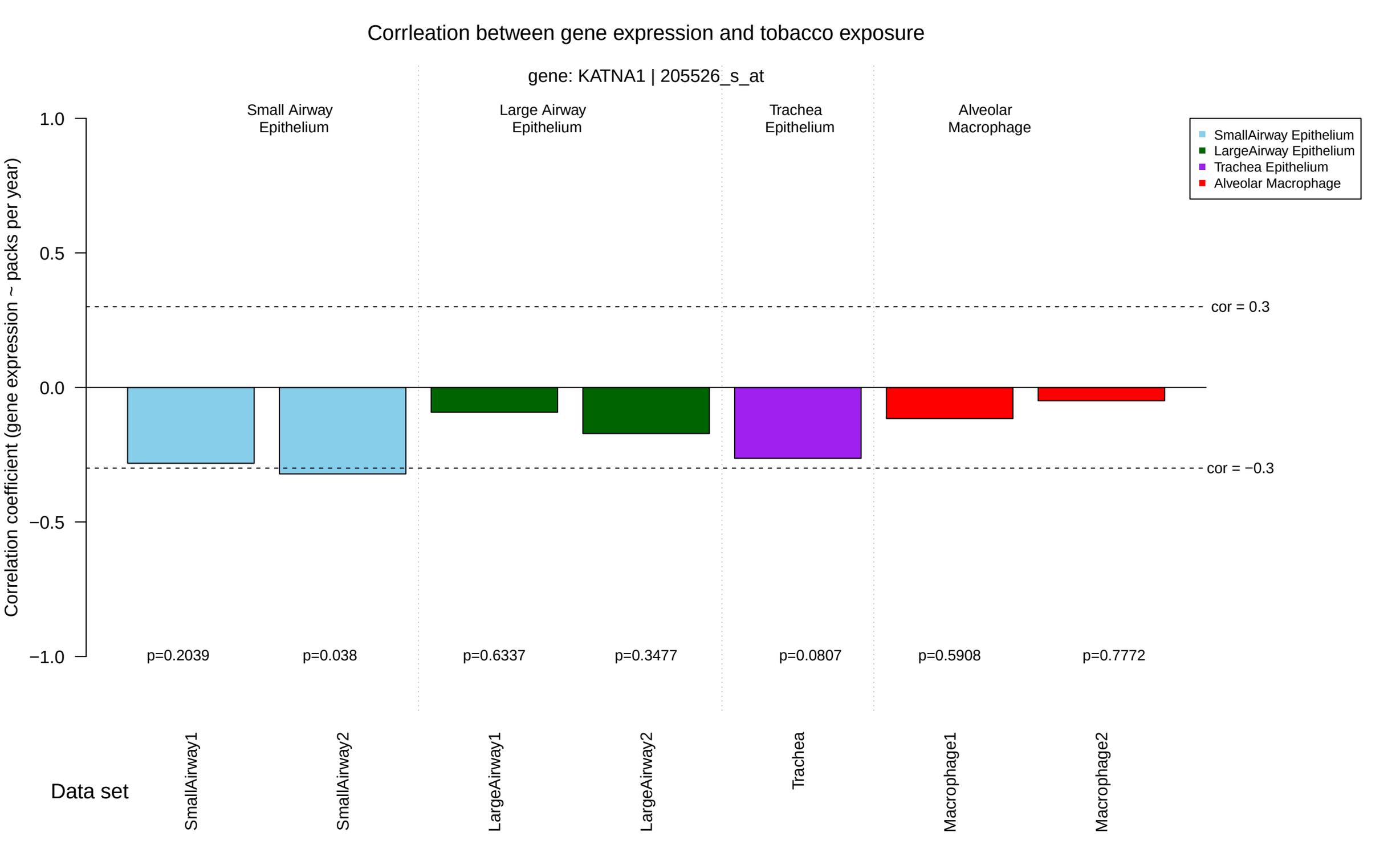

set: LargeAirway1 (p=0.013)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2823)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0153)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9794)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9828)
KATNA1 | 205526_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KATNA1 | 205526_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2039
p=0.038
p=0.6337
p=0.3477
p=0.0807
p=0.5908
p=0.7772
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2